

## Computer Networking Fundamentals

**1. In the OSI model, which layer is responsible for segmenting data and providing reliable end-to-end communication?**

- A. Session Layer
- B. Transport Layer
- C. Network Layer
- D. Presentation Layer

**Answer:** B

**AI Analyzing:** The Transport layer (Layer 4) is responsible for end-to-end communication, including segmentation, flow control, and error correction (in the case of TCP).

**2. What is the primary function of the Address Resolution Protocol (ARP)?**

- A. To translate human-readable domain names into IP addresses
- B. To assign dynamic IP addresses to hosts on a network
- C. To map a known IP address to a physical MAC address
- D. To establish a secure encrypted tunnel between two endpoints

**Answer:** C

**AI Analyzing:** ARP is used to find the Media Access Control (MAC) address associated with a specific IP address on a local area network.

**3. Which of the following protocols operates at the Application Layer of the TCP/IP model and is used for secure web communication?**

- A. FTP
- B. SMTP
- C. SNMP
- D. HTTPS

**Answer:** D

**AI Analyzing:** HTTPS (Hypertext Transfer Protocol Secure) is an application layer protocol that uses TLS/SSL to encrypt communication between a client and a web server.

**4. What is the main difference between a Hub and a Switch in a local area network (LAN)?**

- **A.** A switch broadcasts data to all ports, while a hub sends data to a specific port
- **B.** A hub broadcasts incoming data to all ports, whereas a switch forwards data only to the destination port based on MAC addresses
- **C.** A hub operates at the Network Layer, while a switch operates at the Physical Layer
- **D.** A switch is a passive device, while a hub is an active device that requires power

**Answer:** B

**AI Analyzing:** A hub is a basic device that transmits received data to all connected devices. A switch is more intelligent; it learns MAC addresses and directs traffic only to the specific intended recipient, improving efficiency.

**5. Which field in the IPv4 header is used to prevent packets from circulating indefinitely in a network loop?**

- **A.** Header Checksum
- **B.** Protocol
- **C.** Fragment Offset
- **D.** Time to Live (TTL)

**Answer:** D

**AI Analyzing:** The Time to Live (TTL) field is decremented by 1 by every router that processes the packet. If the TTL reaches zero, the packet is discarded, preventing it from looping forever.